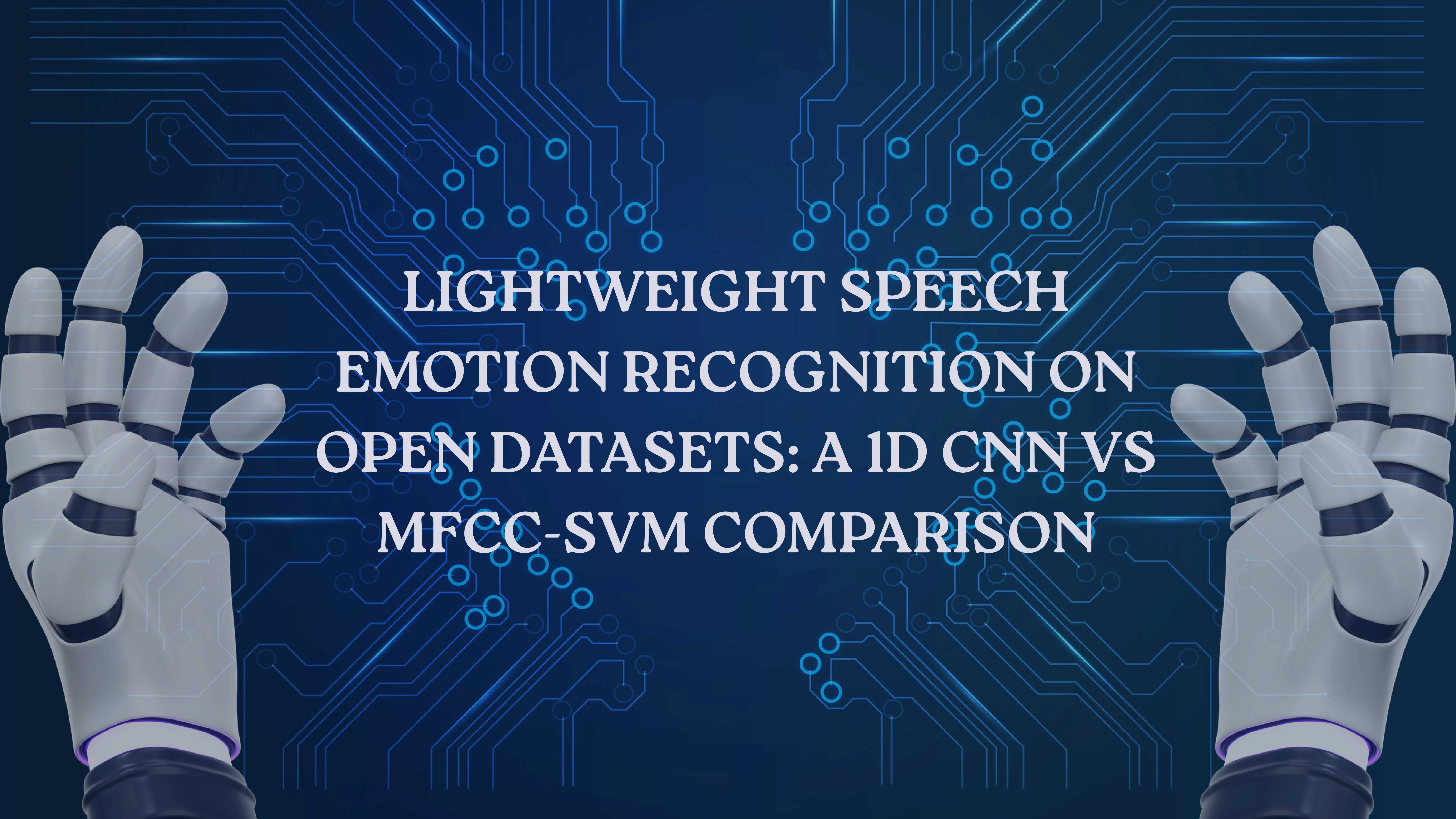
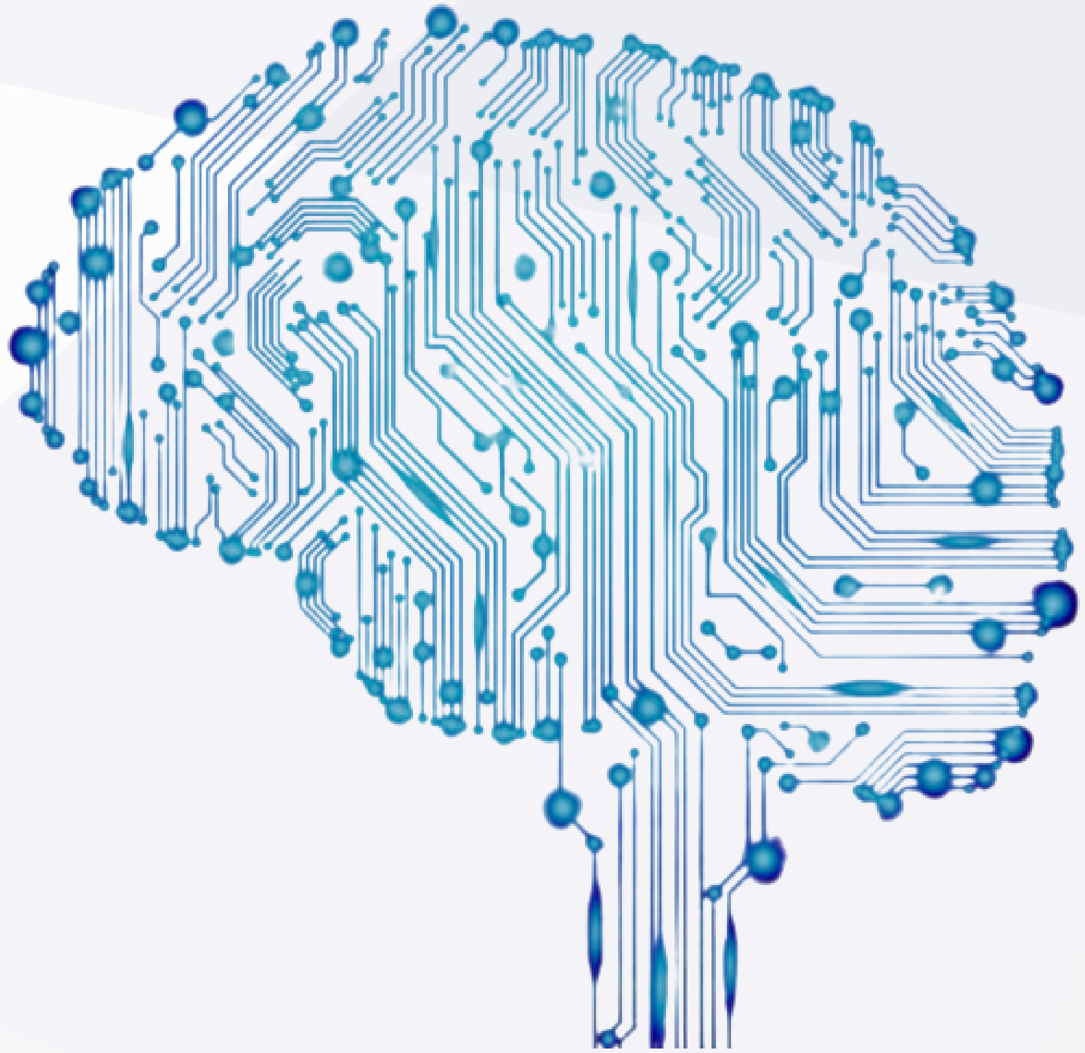


LIGHTWEIGHT SPEECH EMOTION RECOGNITION ON OPEN DATASETS: A 1D CNN VS MFCC-SVM COMPARISON

WHAT IS SER AND WHY IT MATTERS

- Speech carries paralinguistic info: pitch, tone, energy reveal emotion
- Applications: affect aware assistants, healthcare screening, call center routing, adaptive games
- Must run on cpu (laptops, edge devices) no gpu assumed



PROBLEM STATEMENT

01. **Data Input:** 4s audio clip of acted english speech

02. **Output** 1 of 6 emotions (neutral, calm, happy, sad, angry, fearful)

03. **constraints** generalize to unseen speakers, infer under 1 ms/sample on cpu

04. **two approaches** mfcc+svm baseline vs 1d cnn on log-mel-spectrograms

OUR APPROACH AT A GLANCE

01. ravdess dataset, speaker -disjoint split (actors 1-19/20-22/23-24)

02. svm: gridsearched rbf kernel, c capped at 1 (regularization finding)

03. 1d cnn: 132k params, 3 conv1d blocks, log-mel input

04. evaluation: test accuracy, per class recall, latency, confusion matrices

CONTRIBUTIONS

01. clean reproducible 8 notebook pipeline with strict speaker disjoint split

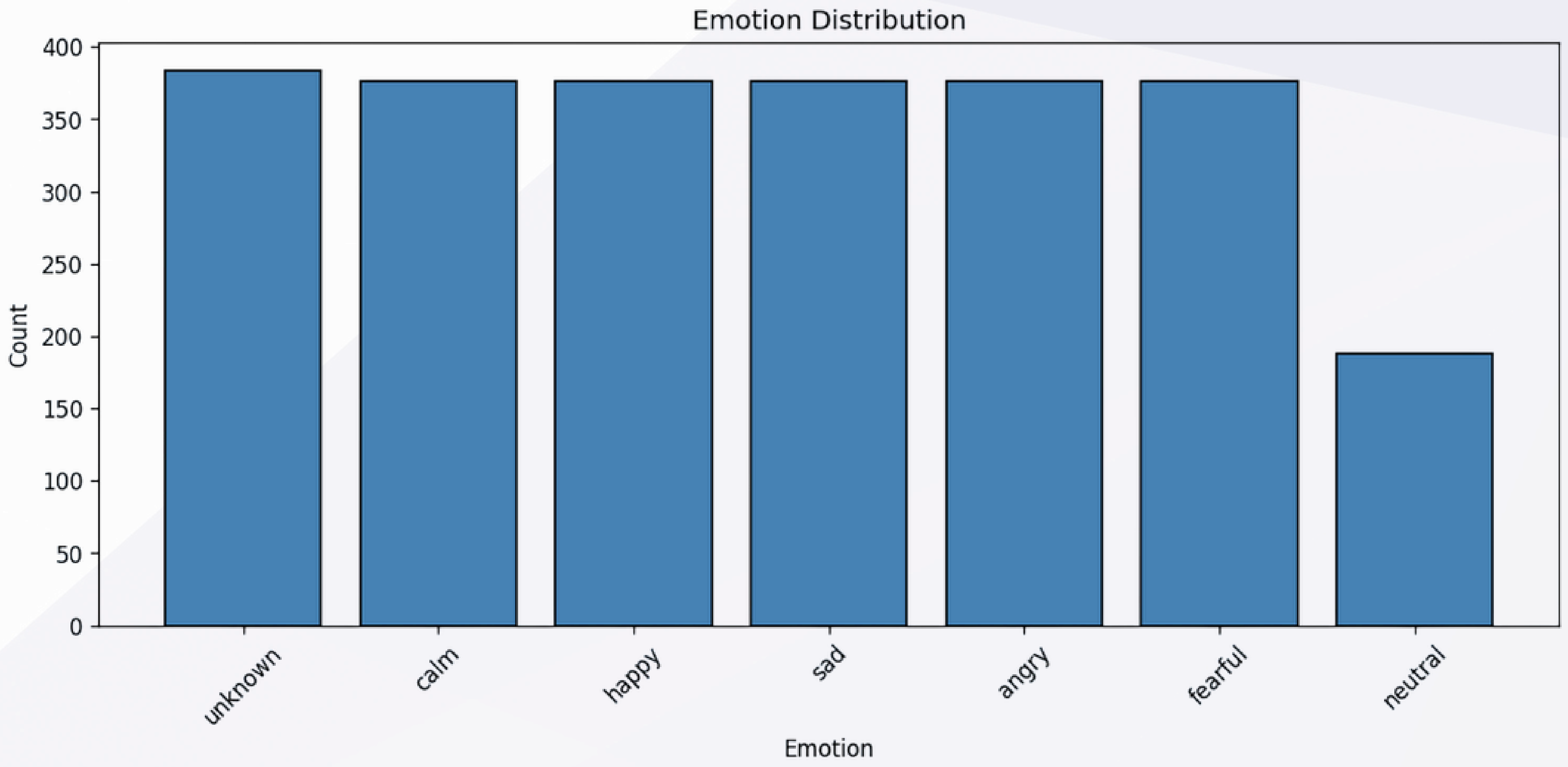
02. regularization analysis: capping svm c reduces train test gap 35pp -> 24pp, improves test accuracy +5pp

03. per emotion error analysis: fearful easiest (90.6%), neutral hardest (50%)

04. cpu latency benchmarks: both under 1 ms/sample

RAVDESS DATASET

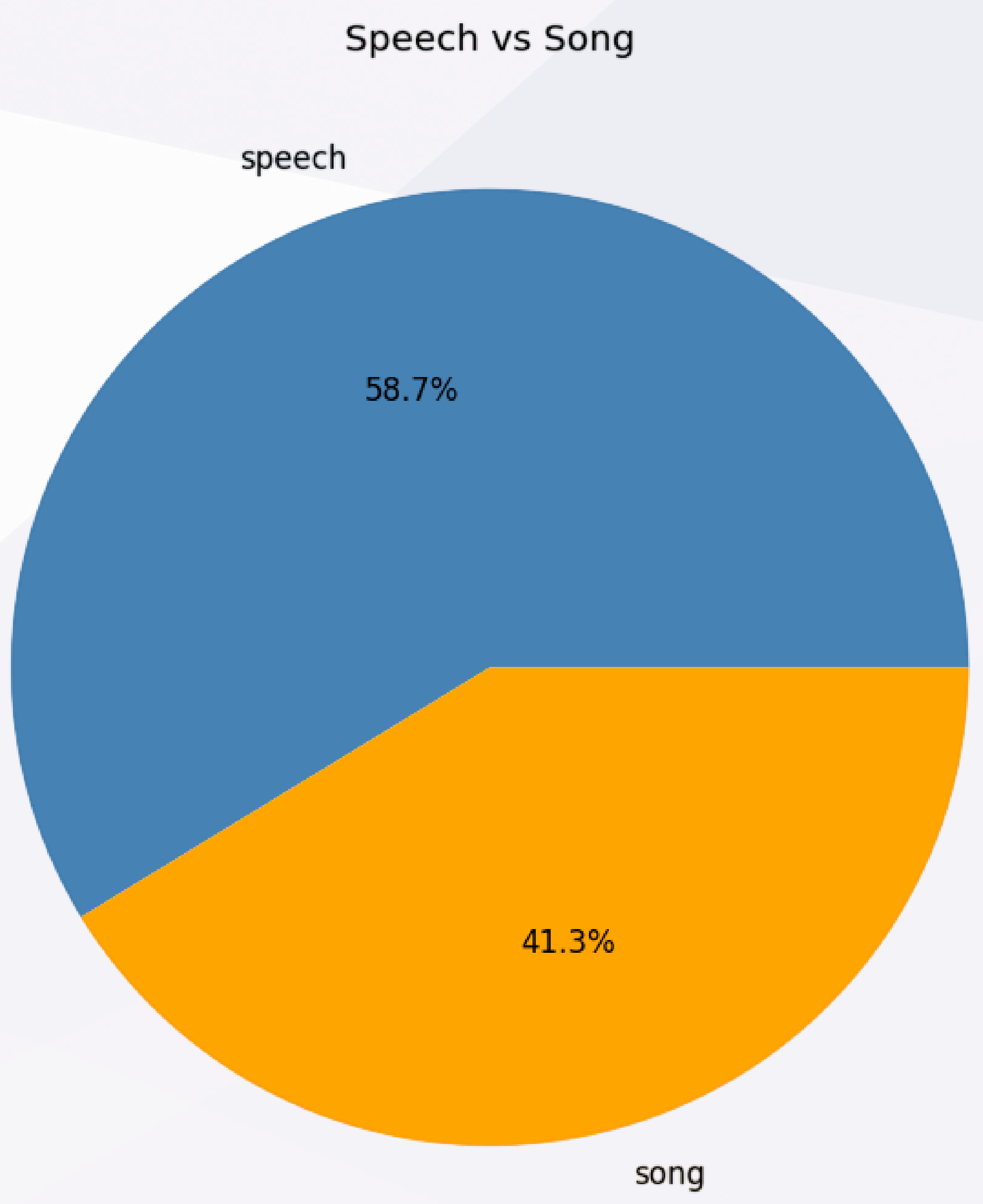
- 24 north american english actors (12 male, 12 female
- speech + song, 2 statements each, normal/strong intensity
- full: 7356 files, 8 emotions
- 1628 train / 264 val / 176 test
- our 6 class subset: neutral, calm, happy, sad, angry, fearful (drop disgust, surprise)



- neutral under represented (~188 vs ~376 per other emotion)

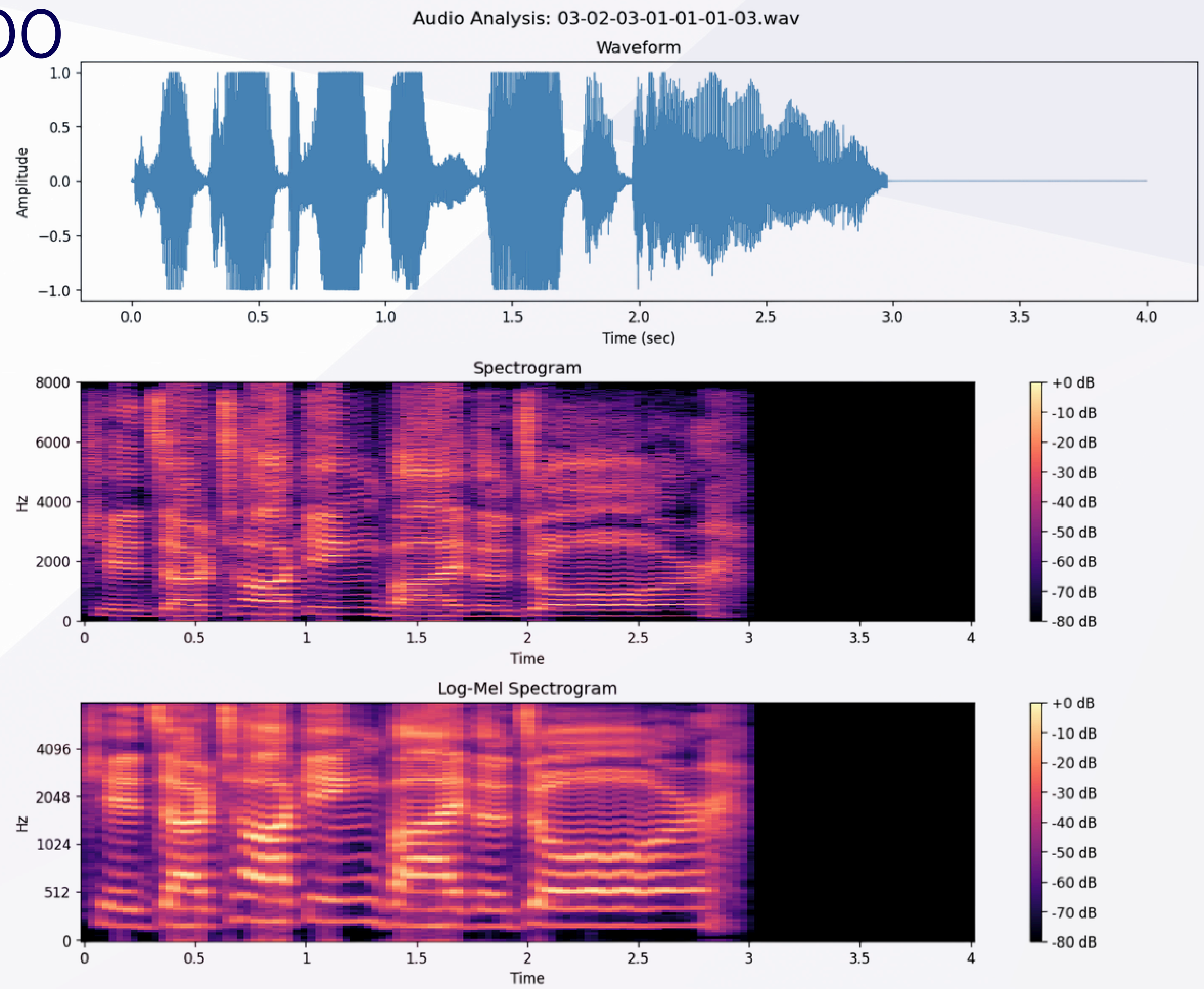
SPEAKER-DISJOINT SPLIT

- train: actors 1-19 (1628 samples)
- val: actors 20-22 (264 samples)
- test: actors 23-24 (176 samples)
- no speaker overlap between splits measures true cross speaker generalization



PREPROCESSING

- resample 16khz mono, trim/pad 4s (64000 samples
- silence trimming (-25db threshold), pre emphasis (0.97), rms normalization
- for cnn: 128 band log-mel-spectrogram, 251 frames, per sample normalized
- for svm: 40 mfcc + delta + delta-delta, mean+std pooled -> 240 dim vector



PREPROCESSING

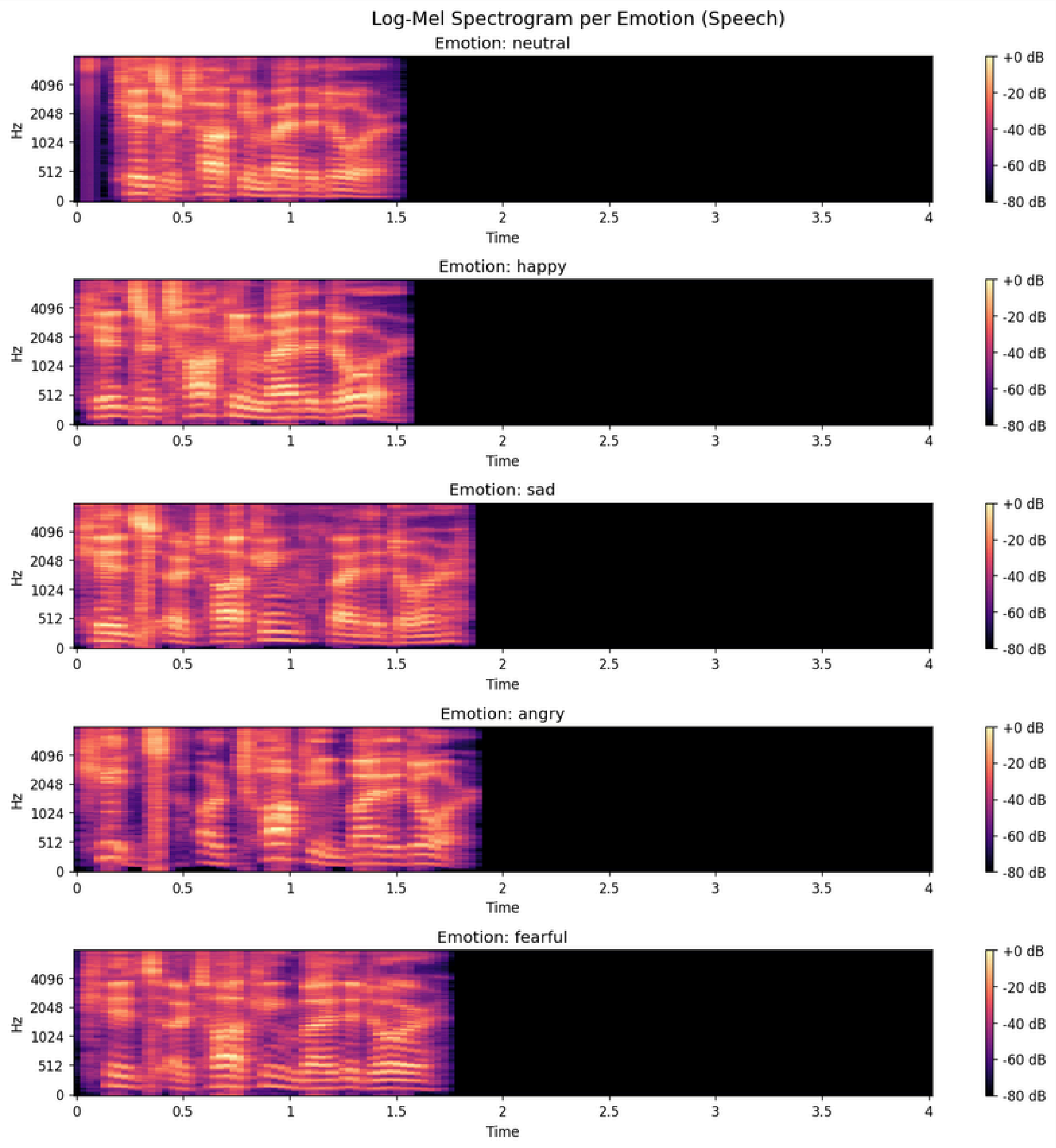


FIG : SAMPLE SPECTROGRAMS 6 PANELS, ONE SPECTROGRAM PER EMOTION, SHOWS DISTINGUISHABLE ACOUSTIC PATTERNS

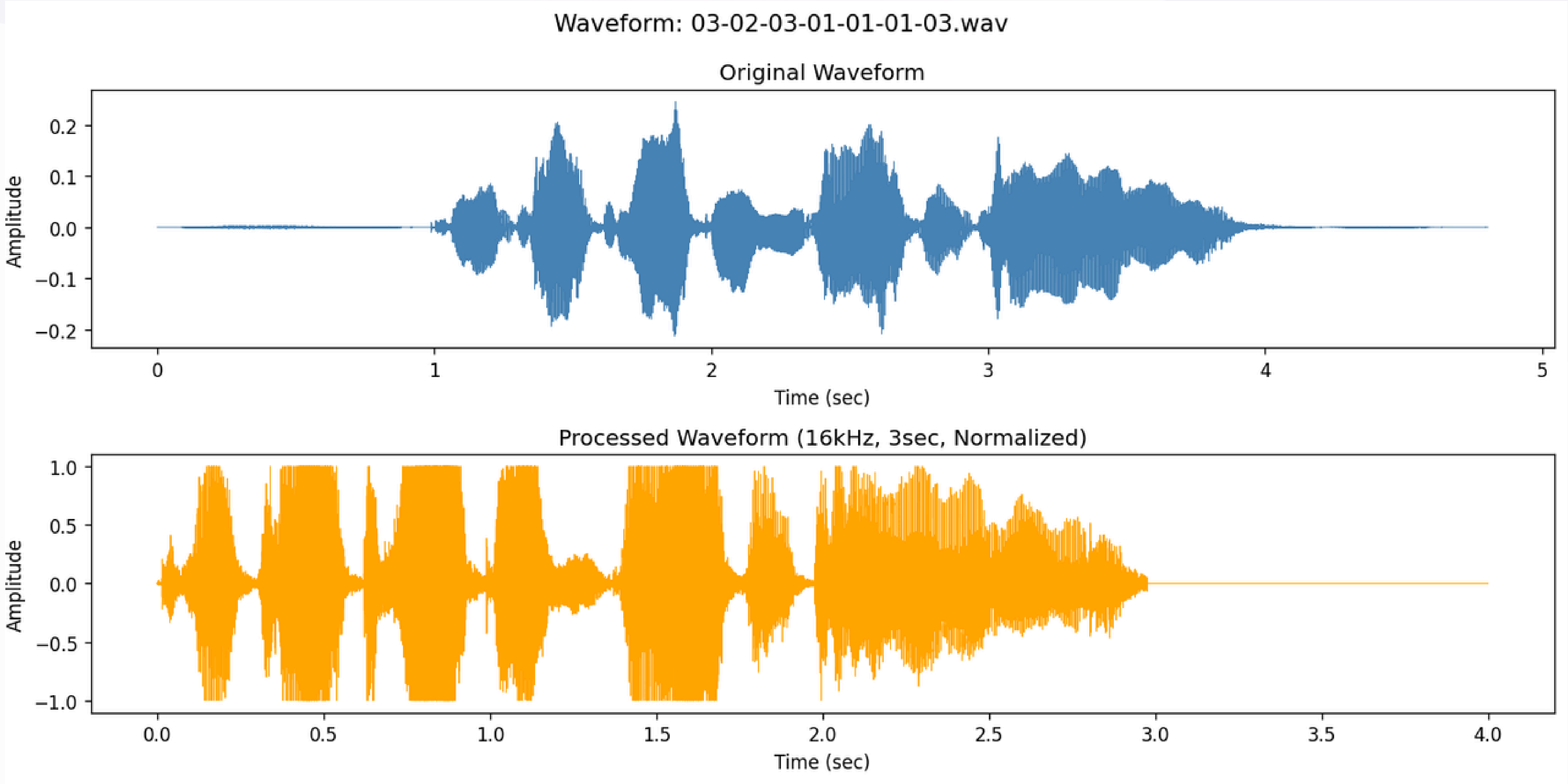


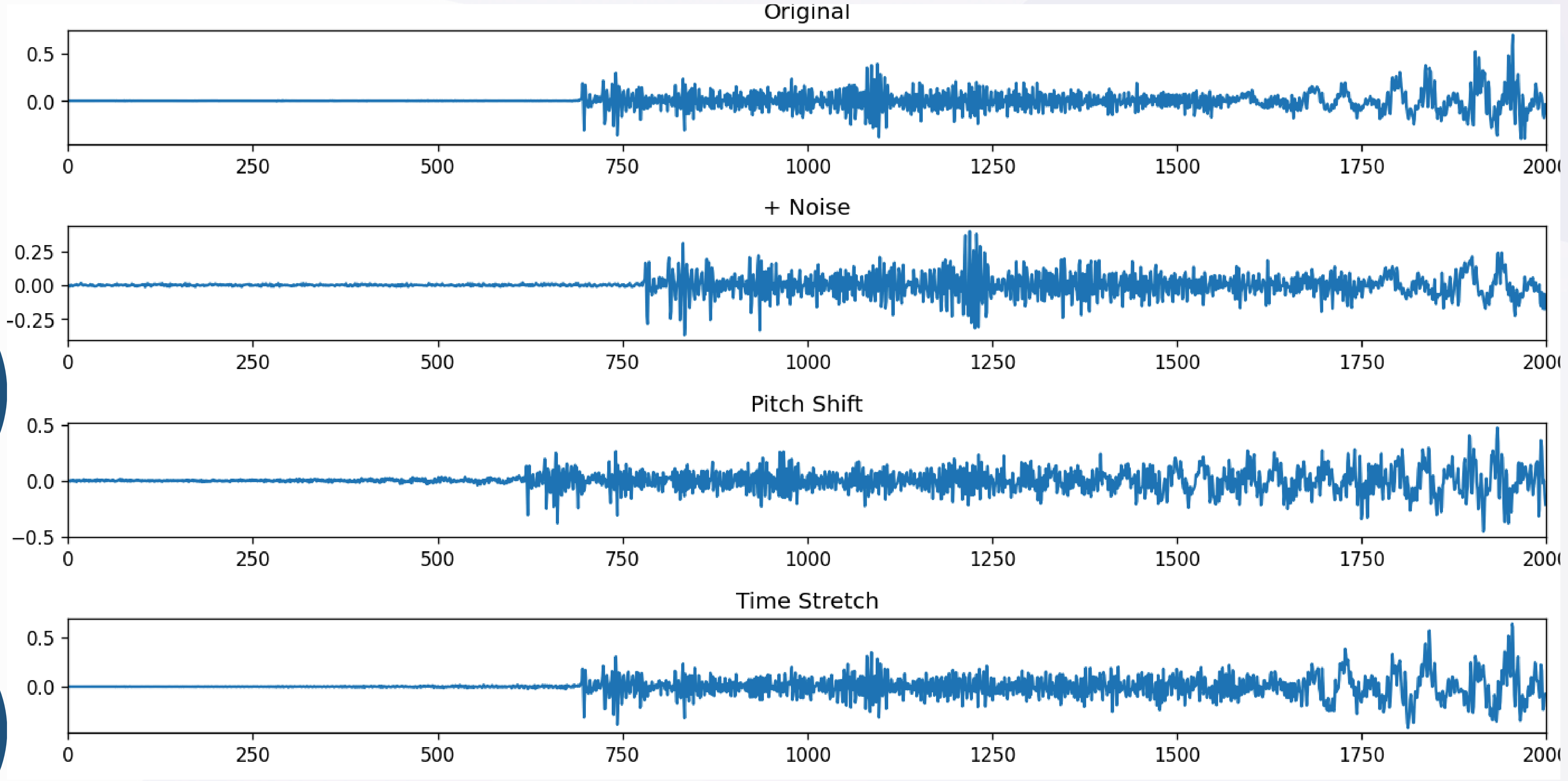
FIG : WAVEFORM COMPARISON ORIGINAL VS PROCESSED: 16KHZ, 4S, NORMALIZED AMPLITUDE

FEATURE EXTRACTION & AUGMENTATION

01. log-mel: 128 bands, 1024-pt fft, 256 sample hop, 251 frames

02. mfcc: 40 coeffs + delta + delta delta, pooled -> 240 dim

03. specaugment: time mask 30, freq mask 12, n=2 (during cnn training only)



SVM BASELINE

- standardscaler on 240 dim mfcc features
- 5 fold grouped cv grid search (15 combos x 5 folds = 75 fits)
- kernels: rbf, linear, polynomial degree 2
- key finding: initial c in {0.1, 1, 10, 50} overfit (100% train, 65% test)
- capping c at {0.1, 1} reduced gap 35pp -> 24pp, test accuracy 65% -> 70.45%
- final: rbf kernel, c=1.0, gamma="scale", probability=true
- training: ~2 min on cpu

SVM BASELINE HYPERPARAMETERS.

parameter	value
feature scaler	standardscaler
kernel	rbf
c	1.0
gamma	"scale"
probability	true
cv folds	5 (grouped by actor)
grid size	15 combos × 5 folds = 75 fits

1D CNN ARCHITECTURE

01. input: (128, 251) log-mel-spectrogram

02. conv1d(128->64, k=5) + bn + relu + maxpool(2)

03. conv1d(64->128, k=5) + bn + relu + maxpool(2)

04. onv1d(128->128, k=3) + bn + relu + adaptiveavgpool(1)

05. dropout(0.3) + linear(128, 6)

06. 132,806 parameters (~528 kb on disk)

CNN TRAINING

- loss: cross entropy with label smoothing 0.05
- optimizer: adamw (lr=1e-3, weight_decay =5e-4)
- scheduler: onecyclelr (max_lr=3e-3, 5-epoch warmup)
- batch size: 32, epochs: 80 (early stopping patience 20)
- device: gpu for training (~47 s for 80 epochs), cpu for inference
- specaugment during training only

1D CNN TRAINING HYPERPARAMETERS.

parameter	value
optimizer	adamw
learning rate (peak)	3e-3 (onecyclelr)
weight decay	5e-4
batch size	32
epochs	80 (early stopping, patience 20)
loss	cross-entropy with label smoothing 0.05
specaugment	time mask=30, freq mask=12, n=2
dropout	0.3
parameters	132,806

EVALUATION PROTOCOL

01. metrics: train/val/test accuracy, per class recall, confusion matrix, cpu latency

02. test set held out from all training + hyperparameter selection

03. latency: average 50 (svm) or 200 (cnn) inference calls after 5 call warmup

QUANTITATIVE COMPARISON

TRAIN/VALIDATION/TEST ACCURACY FOR BOTH MODELS.

model	train	val	test	gap	params
svm rbf	94.47%	62.88%	**70.45%**	24pp	—
1d cnn	94.90%	**75.38%**	66.48%	28pp	132,806

SVM WINS TEST (+3.97PP); CNN WINS VAL (+12.5PP)

CNN OVERFITS MORE (LARGER TRAIN TEST GAP)

SMALL DATASET (1628 SAMPLES) FAVORS HIGH BIAS SVM

PER-CLASS BREAKDOWN

SVM: EXCELS ON FEARFUL (90.6%) AND CALM (81.2%)

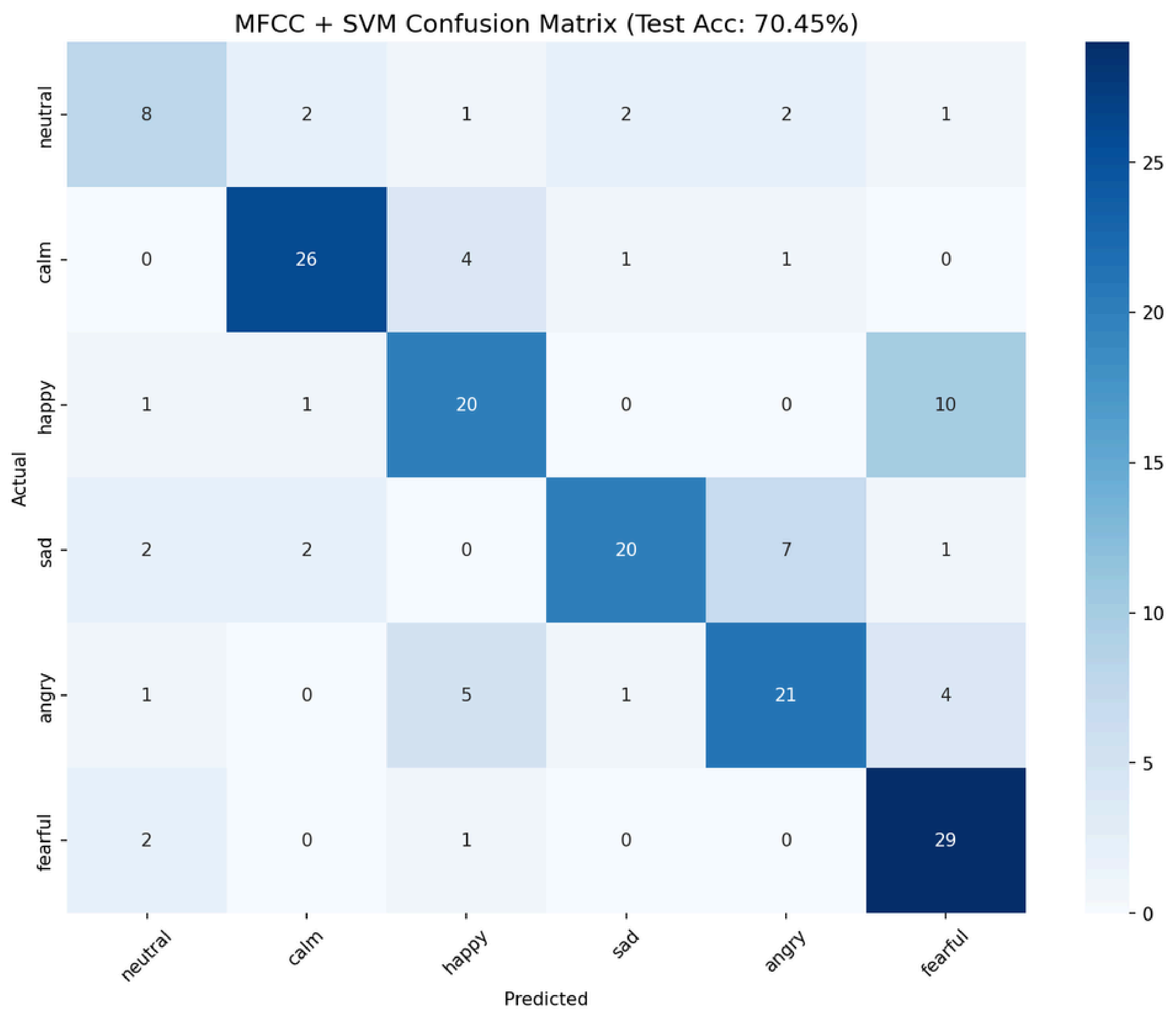
CNN: EXCELS ON ANGRY (84.4%) AND SAD (75.0%)

BOTH: STRUGGLE WITH NEUTRAL (50%) SMALL SAMPLE, ACOUSTIC SIMILARITY TO CALM

PER-EMOTION TEST ACCURACY.

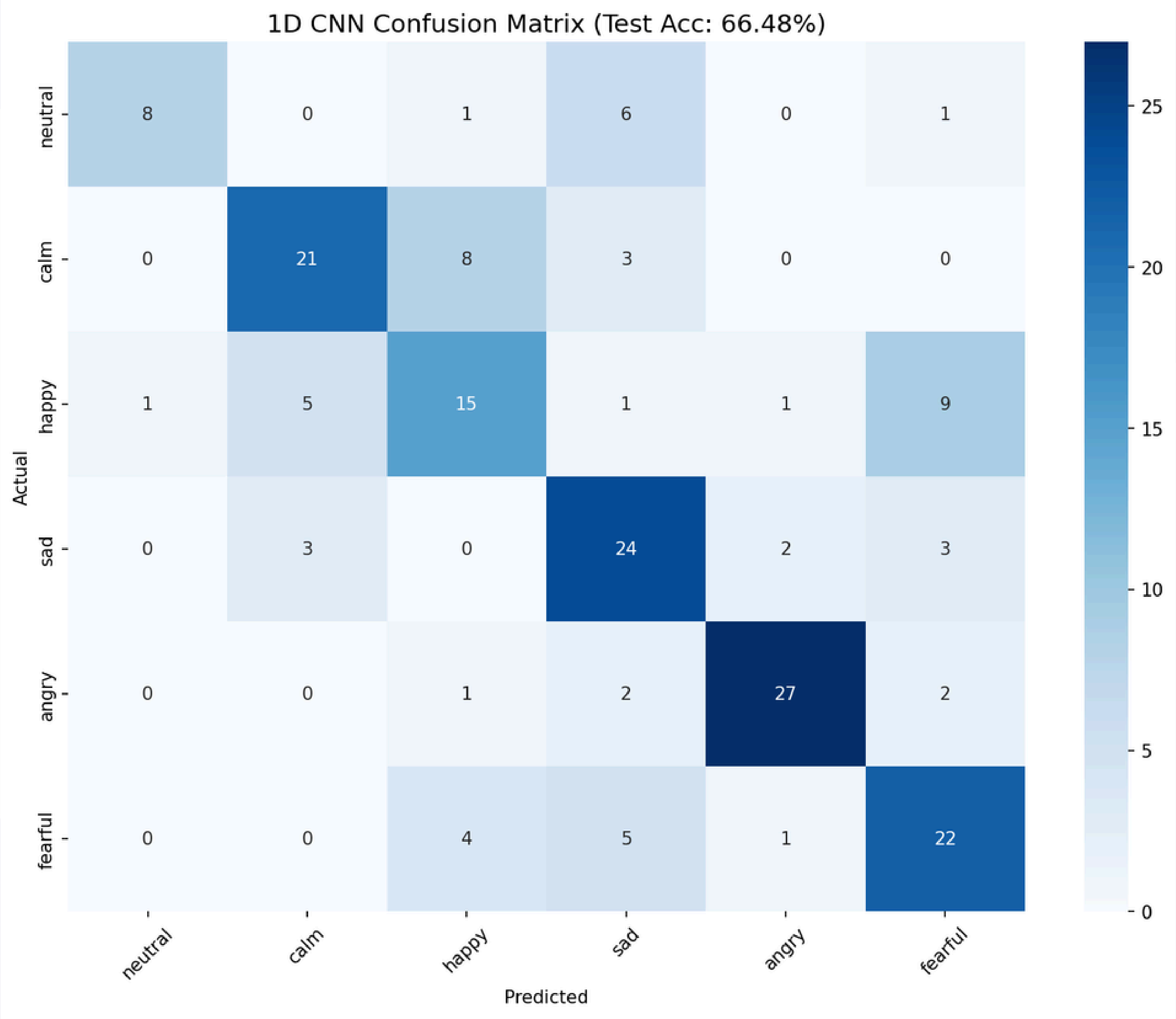
emotion	support	svm rbf	1d cnn
neutral	16	50.0%	50.0%
calm	32	**81.2%**	65.6%
happy	32	62.5%	46.9%
sad	32	62.5%	**75.0%**
angry	32	65.6%	**84.4%**
fearful	32	**90.6%**	68.8%

CONFUSION MATRICES



	neutral	calm	happy	sad	angry	fearful
neutral	8	2	1	2	2	1
calm	0	26	4	1	1	0
happy	1	1	20	0	0	10
sad	2	2	0	20	7	1
angry	1	0	5	1	21	4
fearful	2	0	1	0	0	29

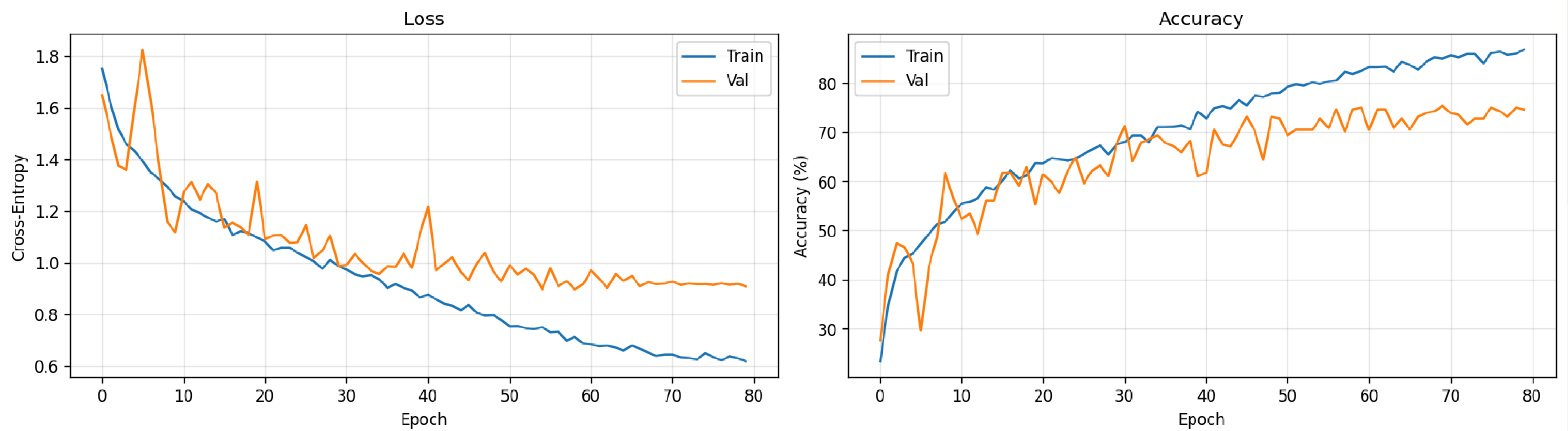
SVM TEST CONFUSION MATRIX HAPPY->FEARFUL MOST COMMON (10/32), BOTH HIGH-AROUSAL*



	neutral	calm	happy	sad	angry	fearful
neutral	8	0	1	6	0	1
calm	0	21	8	3	0	0
happy	1	5	15	1	1	9
sad	0	3	0	24	2	3
angry	0	0	1	2	27	2
fearful	0	0	4	5	1	22

CNN TEST CONFUSION MATRIX HAPPY->CALM, FEARFUL->SAD, CONFUSES AROUSAL LEVELS*

TRAINING CURVE



CNN TRAINING CURVES :
TRAIN ~95%
VAL PLATEAUS ~75%
EARLY STOPPING AT EPOCH 67

EFFICIENCY

CPU INFERENCE LATENCY (PER SAMPLE).

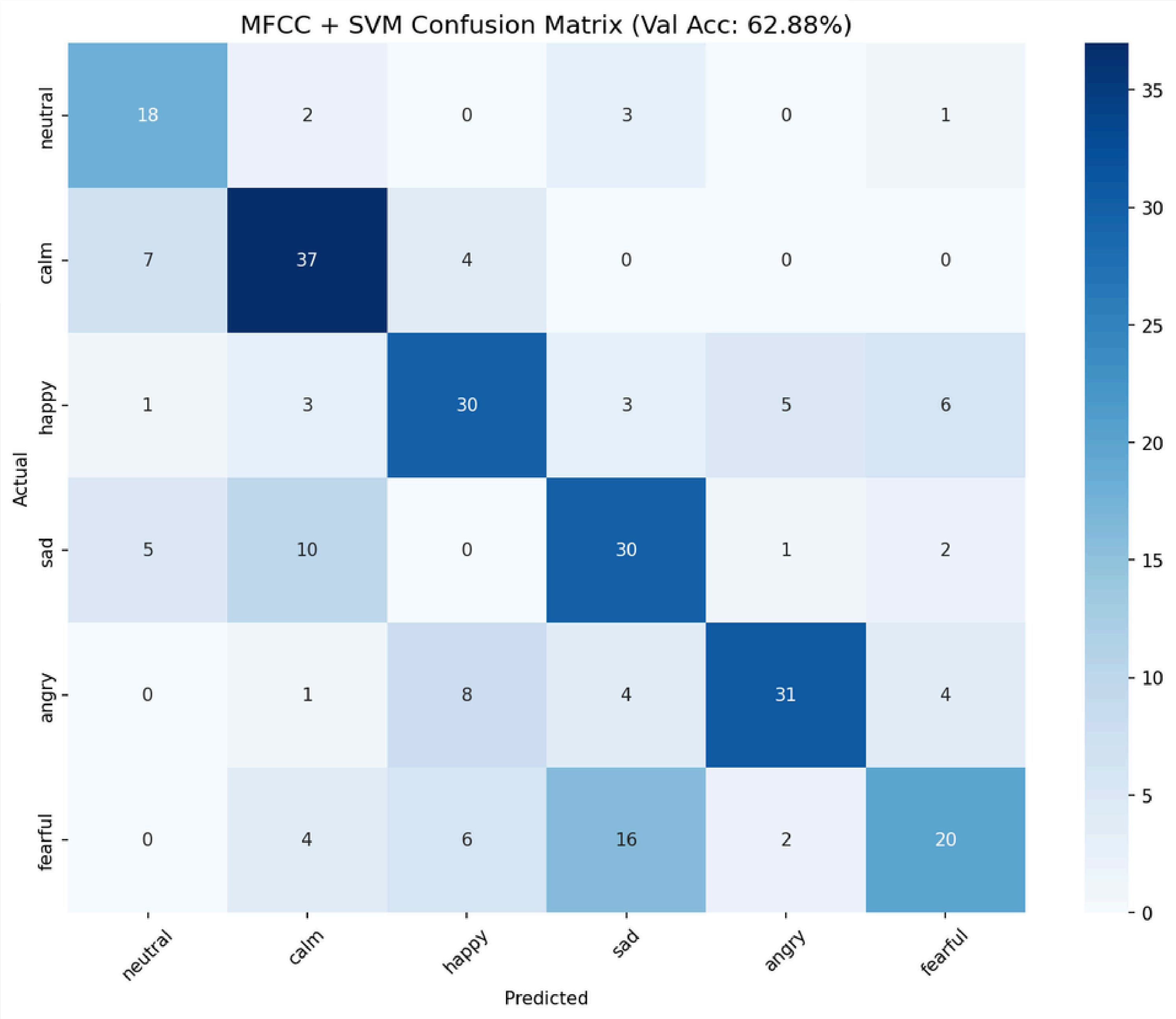
model	mean	p50	p99	throughput
svm rbf	0.195 ms	0.180 ms	0.331 ms	5,125 samp/s
1d cnn	0.530 ms	0.500 ms	0.863 ms	1,888 samp/s

- **SVM 2.7X FASTER**
- **SVM OVERHEAD: KERNEL EVAL ON 1628 SUPPORT VECTORS**
- **BOTH UNDER 1 MS TARGET**
- **CNN OVERHEAD: CONV1D OPS**

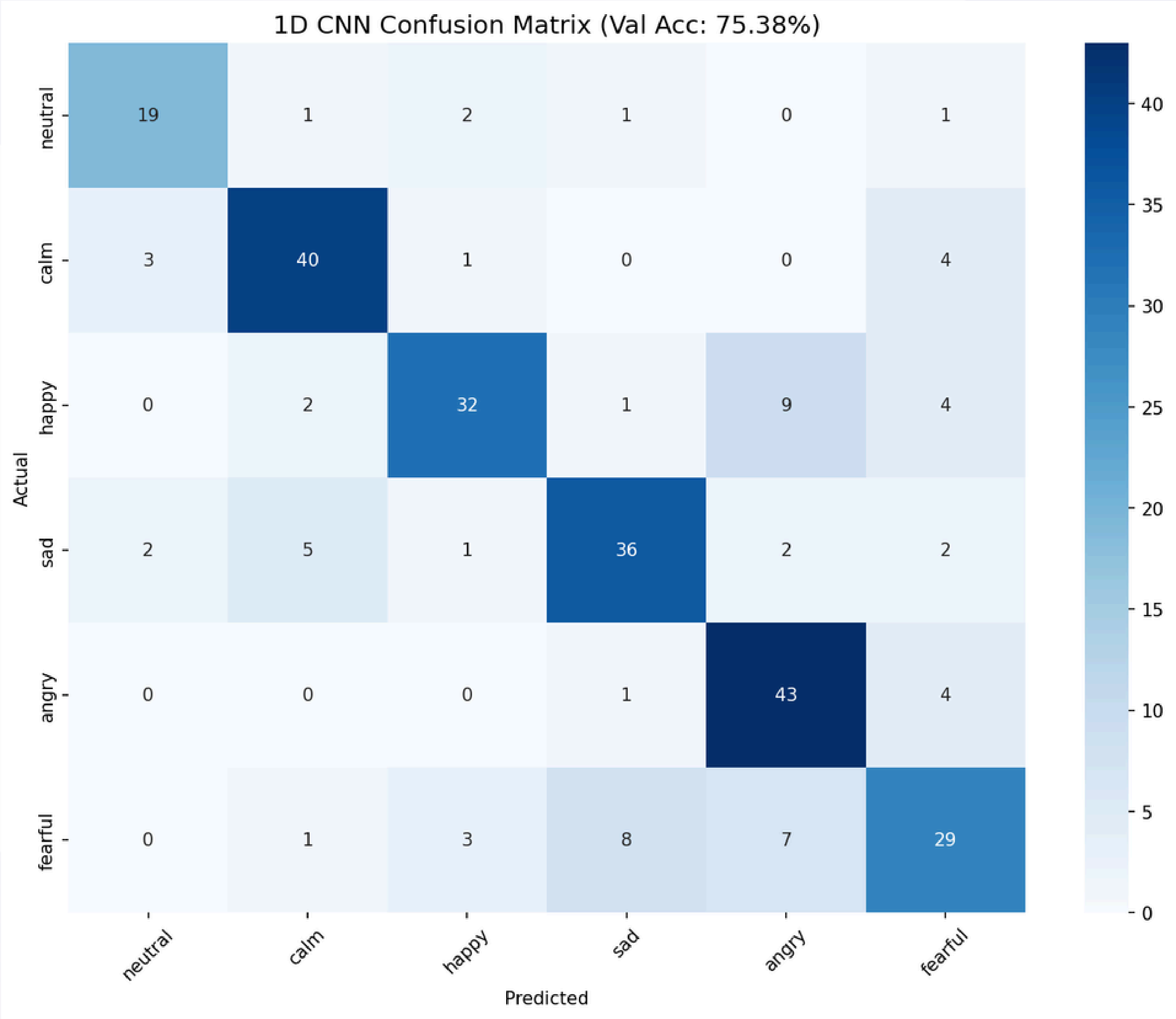
WHY SVM WINS

01.	bias-variance tradeoff	628 samples favors high bias svm
02.	feature representation	mfcc+delta is strong hand crafted feature for emotion
03.	kernel capacity	rbf + c=1 is moderate capacity capping c was key regularization fix
04.	literature range	55-80% for cross speaker ravdess our 70.45% svm is competitive

VALIDATION CONFUSION MATRICES



svm validation confusion matrix
same patterns as test



cnn validation confusion matrix
broader confusion across arousal levels

LIMITATIONS

01. SMALL DATASET — 1628 TRAINING SAMPLES IS SMALL FOR DEEP LEARNING

02. ACTED EMOTIONS — RAVDESS ACTED EMOTIONS ARE EXAGGERATED VS NATURAL SPEECH

03. SINGLE DATASET — CROSS-DATASET GENERALIZATION UNTESTED (RAVDESS -> CREMA-D, IEMOCAP)

04. 6 CLASS SUBSET — DROPPED DISGUST AND SURPRISE LIMITS COMPARISON TO 8-CLASS RESULTS

05. CPU-ONLY — NO GPU LATENCY MEASUREMENTS (WOULD BE 10-100X FASTER)

FUTURE WORK

01. RANSFER LEARNING	FINE-TUNE WAV2VEC 2.0 OR HUBERT EXPECTED +10-15PP IMPROVEMENT
02. DATA AUGMENTATION	MIXUP, STRONGER SPECAUGMENT, VOCAL TRACT LENGTH NORMALIZATION
03. MULTI DATASET TRAINING	COMBINE RAVDESS + CREMA-D + IEMOCAP + TESS
04. ATTENTION	SELF-ATTENTION AFTER CONV1D BLOCKS FOR LONG-RANGE DEPENDENCIES
05. QUANTIZATION	INT8 QUANTIZATION COULD REDUCE CNN LATENCY 4X, SIZE 4X

CONCLUSION

- clean, reproducible comparison of svm vs 1d cnn on ravdess
- svm achieves 70.45% test accuracy, 0.19 ms/sample wins on this dataset
- capping svm c was key regularization insight (+5pp test accuracy)
- both models meet sub 1ms cpu latency target
- full open source pipeline: 8 notebooks, typing, configs, docs
- code: https://github.com/elsayedelmandoh/lightweight-speech-emotion-recognition-on-open-datasets

Q & A

Q1: WHY NOT USE A PRE-TRAINED MODEL LIKE WAV2VEC 2.0?

Q2: WOULD DATA AUGMENTATION HELP THE CNN MORE?

Q3: WHY 6 CLASSES INSTEAD OF 8?

Q4: HOW DID YOU CHOOSE THE CNN ARCHITECTURE?

Q5: IS THE 24-28PP TRAIN-TEST GAP NORMAL?